

# Simulation Verification

Project 5 | Original XSim evidence and additional RTL trace

The original uploaded Vivado 2025.1.1 batch output is preserved as Simulation/transcript.txt and Results/run\_20261008\_121952\_125/vivado\_batch.log. Separate original suite PASS files are included. These are simulation/build outputs, not reconstructed physical test data.

## Verification coverage

| Suite             | Verified coverage                                          |
|-------------------|------------------------------------------------------------|
| Cores             | 131,072 results/core; full hidden/scores/class/margin/tags |
| Arithmetic        | 65,536 popcounts; 4,352 score extremes                     |
| Diagnostics       | 1,024 model images; 4,096 checks                           |
| Configuration     | 1,216 checks; occupancies 0..21                            |
| Safety            | 26 directed checks                                         |
| Integrated system | 6,615 checks; 1,250 raw comparisons                        |

## Measured cycle behavior

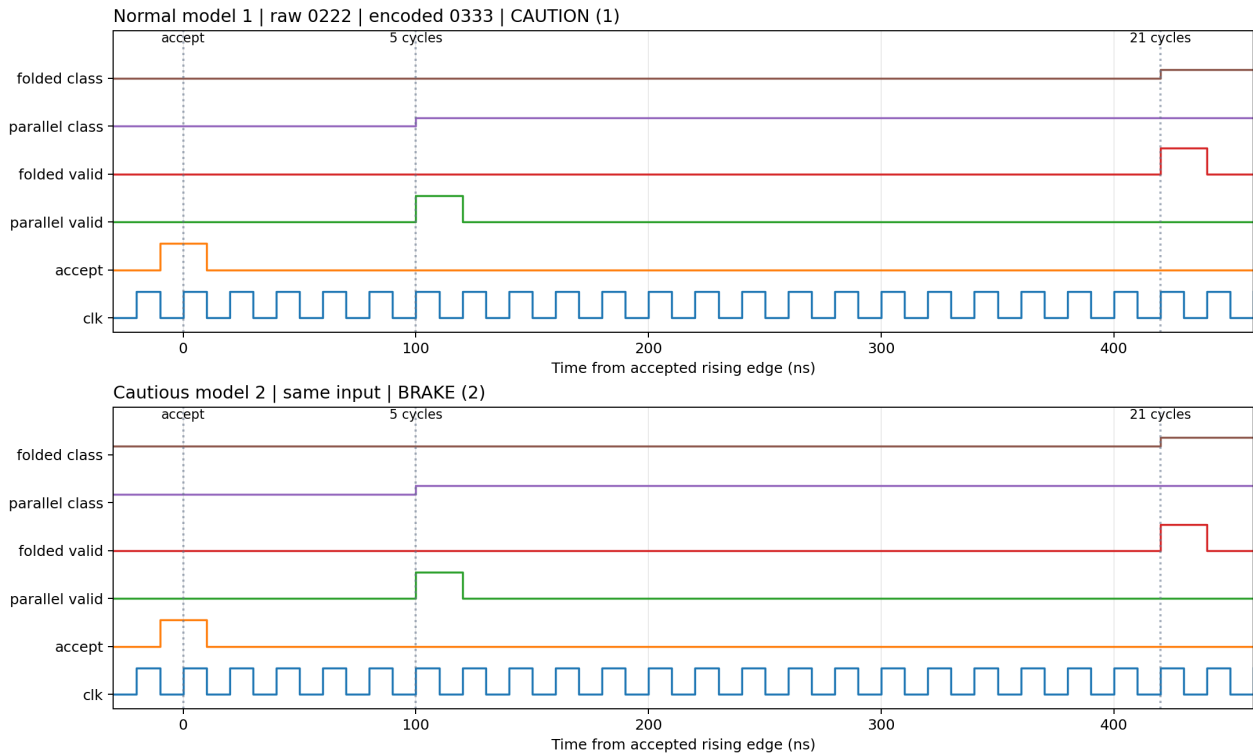
Parallel result latency: 5 cycles; initiation interval: 1 cycle. Folded latency: 21 cycles; initiation interval: 22 cycles. At the implemented 50 MHz clock these neural result latencies are 100 ns and 420 ns. Host/JTAG and frame-sampling delays are separate.

## Synthetic policy agreement

Both profiles classify 125/125 held-out synthetic cases correctly. 625/625 full-domain checks include training cases. Exhaustive 16-bit arithmetic vectors validate exact RTL/reference agreement; many are not legal four-feature physical inputs. This does not measure real-world sensor safety accuracy.

## Actual RTL waveform

An additional Icarus run executes Testbench/tb\_waveform.sv against the committed RTL, model words and independent frozen goldens. Twelve cases (six/profile) check hidden state, all signed scores, class, margin, tag and exact 5/21-cycle latency.



Transitions plotted from waveform.vcd. The two panels show raw input 0222, encoded 0333, under model IDs 1 and 2. Clock period is 20 ns. CSV observations sample 2 ns after edges; cycle differences are unchanged.

## Reproduction and provenance

Simulation/README.md provides commands. Python/run\_open\_source\_tests.py reruns all five primary suites with Icarus; RUN\_PROJECT5.cmd reruns them with XSim and builds the design.

Testbench/tb\_waveform.sv and Python/plot\_waveform.py regenerate the additional trace and PNG. Icarus, Python and optional matplotlib are needed for that plotting path.

Physical verification is documented separately in the project report. Missing original hardware core CSV/PASS files are not substituted with simulated data.